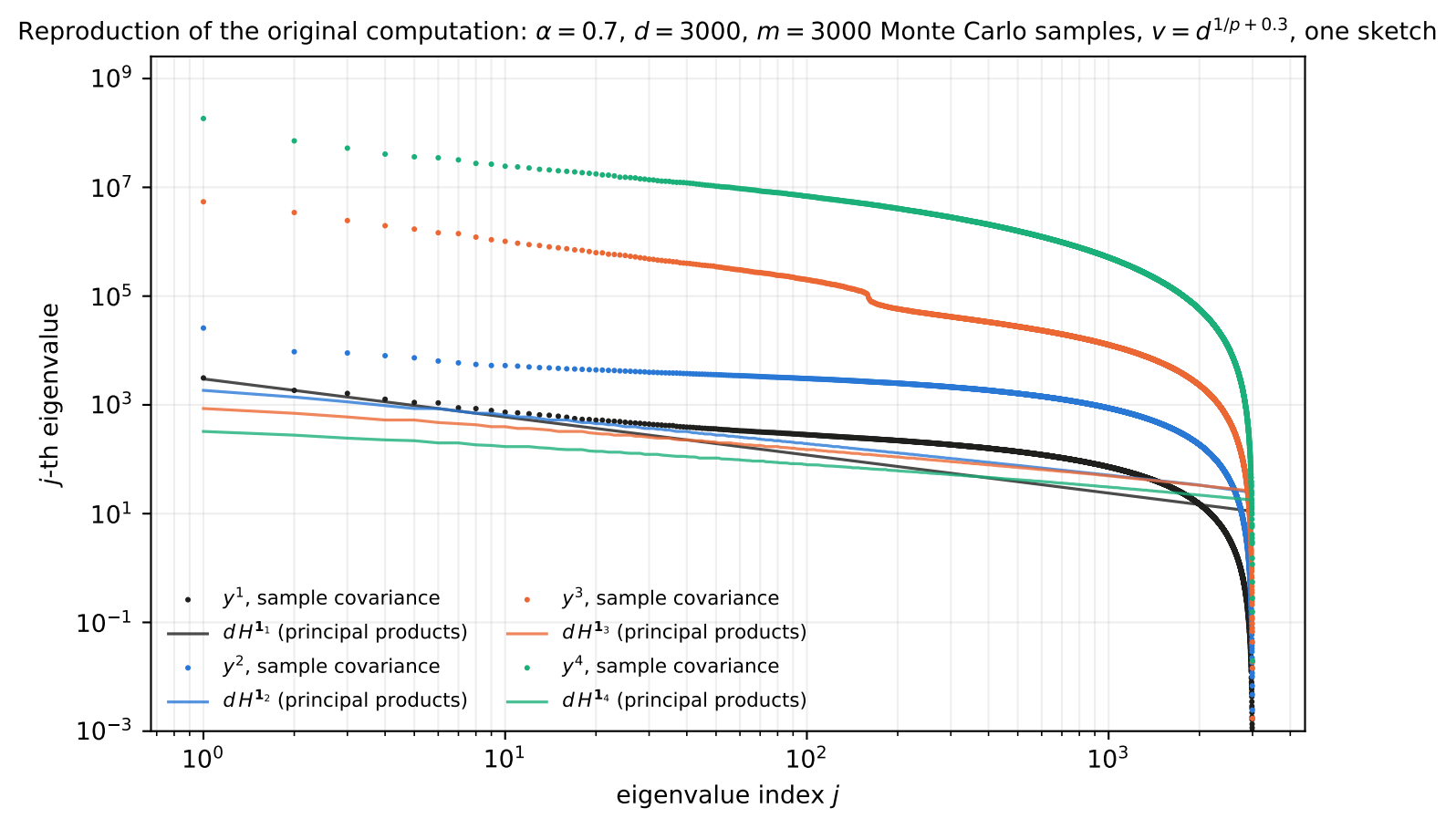

Reproduction of the original computation: $\alpha = 0.7$, $d = 3000$, $m = 3000$ Monte Carlo samples, $v = d^{1/p + 0.3}$, one sketch
j-th eigenvalue
eigenvalue index j
10^{9}
10^{7}
10^{5}
10^{3}
10^{1}
10^{-1}
10^{-3}
10^{0}
10^{1}
10^{2}
10^{3}
y^1, sample covariance
$dH^{\mathbf{1}_1}$ (principal products)
y^2, sample covariance
$dH^{\mathbf{1}_2}$ (principal products)
y^3, sample covariance
$dH^{\mathbf{1}_3}$ (principal products)
y^4, sample covariance
$dH^{\mathbf{1}_4}$ (principal products)